Metadata Management for Cluster File Systems with Predictive Placement and File Grouping

Problem.
Clustered file systems like CephFS are a strong fit for large shared storage, but metadata-heavy workloads are still a pain point. The Ceph docs note that metadata operations can make up more than half of all file-system operations, and CephFS scales metadata by spreading directory subtrees across multiple metadata servers (MDSs) using dynamic subtree partitioning. That helps, but it is still mostly reactive and workloads with many tiny files can overwhelm the metadata path even when data I/O is modest.

The Idea.
I wanted to study two complementary ways to reduce CephFS metadata overhead. The first is a predictive metadata placement policy that tries to identify hot directories early (by tracking per-directory metadata activity) and proactively pin or migrate them before they become bottlenecks. The second is a log-structured small-file ingest layer that reduces metadata pressure by packing many tiny logical files into a smaller number of large append-only segment files, with a separate index for lookup. In related work, in this TableFS paper, the authors showed that by storing metadata and tiny files as key-value pairs in a database, you can optimize data-intensive workloads. I think these ideas will transfer well to CephFS too.

Evaluation plan.
I will run experiments on a small CloudLab Ceph cluster with multiple MDS daemons and several client nodes. The native CephFS with the default balancer and CephFS with static subtree pinning will be the baselines. Workloads could include mdtest and custom scripts that create, stat, and delete large numbers of small files, plus mixed workloads to make sure improvements do not come at the cost of making normal I/O much worse. I especially want to focus on measuring the tradeoff between the extra lookup overhead and less metadata for the small file ingest layer. Standard measures like metadata throughput, p95/p99 latency for create/open/stat, MDS CPU and load distribution, and the overhead of compaction in the overlay will be things to test for.

Challenges.
The main risk in the predictive policy is that hotspots may move too quickly for simple forecasts to help. The main risk in the overlay is preserving file-system semantics without building a full new file system. To keep the project scoped, I will focus on metadata-heavy workloads, and simple correctness first, rather than trying to support every POSIX corner case.

- **75% Goals:** Set up CephFS, reproduce a metadata bottleneck with mdtest, and implement a working predictive policy
- **100% goal**: implement both systems in working form and compare them against native CephFS, with clear graphs for throughput, tail latency, and MDS load balance.
- **125% goal**: Having a small NN that is trained on workloads to predict emerging metadata hotspots from recent access patterns. More than the implementation, I think the thorough evaluation needed here makes it the stretch 125% goal.

References:

"Ceph File System." *Ceph Documentation*, Ceph, docs.ceph.com/en/latest/cephfs/index.html.

Ren, Kai, and Garth Gibson. "TABLEFS: Enhancing Metadata Efficiency in the Local File System." *2013 USENIX Annual Technical Conference (USENIX ATC 13)*, USENIX Association, 2013, pp. 145–156.

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